

Capstone Project

EGYPT FOOD SECURITY MACRO SHOCK ANALYSIS

Guidebook

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About the Authors: A Mosaic of Expertise

We are a team of five brought together by the Digital Egypt Pioneers Initiative (DEPI), united by a shared interest in data analytics and a drive to tackle one of Egypt's most pressing challenges — food security in the face of macroeconomic instability. Our backgrounds span computer science, information systems, and business, giving the project both technical depth and real-world analytical perspective.

Technical Growth & New Perspectives

Industry Expertise & Strategic Insight

One Shared Mission

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Project Timeline Summary (March–June 2026)

The project follows a structured four-month timeline starting in early March 2026 and extending through the end of June. It begins with defining the problem and conducting research, followed by data collection, cleaning, and modeling. After establishing a validated data model, an extended exploratory analysis phase runs through early June to uncover patterns and validate findings.

The visualization phase begins in early June, where the dashboard is designed, implemented, and refined over approximately two weeks. Insights and recommendations are then developed and validated with the team in the penultimate week.

The final session on 22 June is dedicated entirely to delivering the presentation, handling Q&A, and submitting the completed project.

Task	04-03	14-03	24-03	04-04	02-06	15-06	22-06
Problem Definition							
Data Collection							
Data Modeling							
Exploratory Analysis							
Visualization							
Insights & Recommendations							
Presentation & Q&A							

This initiative is offered by the Ministry of Communications and Information Technology under the name Digital Egypt Pioneers Initiative (DEPI), in collaboration with private training partners such as CLS Learning Solutions, Leading 365, Berlitz, and Elharefa. Launched in 2023, the program aims to equip students for the freelance and digital job market, providing specialized tracks in technical skills, soft skills, freelancing, and English.

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Chapter 1: Data Preparation

❖ Overview

❖ Key Steps in Data Preparation Using Power Query

Chapter 1: Data Preparation

Overview

Data preparation is a critical phase in building a reliable and accurate data analysis solution. Before any reporting, modeling, or visualization can occur, the raw data must be cleaned, structured, and transformed to ensure consistency and accuracy. In this project, data preparation was performed using Power Query (M Language), which provides a powerful environment for importing data, cleansing it, and shaping it to support analytical requirements.

In this project, the data preparation work focused on:

- Integrating data from multiple international and national sources into a single, consistent data model covering inflation, exchange rates, food prices, trade volumes, income levels, and household wages.
- Correcting and standardizing information such as dates, country codes, product names, and measurement units across heterogeneous source formats (CSV, Excel).
- Removing errors, null values, and irrelevant historical rows outside the 2020–2024 study window.
- Creating clean dimension and fact tables suitable for star/snowflake schema modeling in Power BI.
- Structuring records so that macroeconomic indicators, food prices, and affordability metrics can be analyzed accurately and cross-filtered.

By completing this preparation, we ensured that all KPIs — including Headline CPI, FX Devaluation YoY%, Average Food Inflation, and the Income Erosion Index — are grounded in trustworthy, well-organized data.

Key Steps in Data Preparation Using Power Query

A standardized transformation approach was applied across all source tables. The following sections document the exact steps used for each query.

1. Table: Inflation (World Bank CPI Data)

Step	Description
1. Import Source File	<pre>Source = Csv.Document(File.Contents("../API_FP.CPI.TOTL.ZG_DS2_en_csv_v2.csv"), [Delimiter=";", Encoding=65001, QuoteStyle=QuoteStyle.None])</pre> Loads the World Bank Consumer Price Index CSV file into Power Query.
2. Promote Headers	<pre>"Promoted Headers" = Table.PromoteHeaders(Source, [PromoteAllScalars=true])</pre> Converts first row into column headers.
3. Remove Top Rows	<pre>"Removed Top Rows" = Table.Skip("#Changed Type", 3)</pre> Skips the 3 metadata rows at the top of the World Bank CSV format.
4. Promote	<pre>"Promoted Headers1" = Table.PromoteHeaders("#Removed Top Rows", [PromoteAllScalars=true])</pre>

Second Headers	Promotes the actual data header row after skipping metadata.
5. Unpivot Year Columns	<pre>#"Unpivoted Other Columns" = Table.UnpivotOtherColumns(..., "Attribute", "Value")</pre> Converts wide year columns (1960-2024) into long format with two columns: Years and Inflation.
6. Rename & Retype	<pre>#"Renamed Columns1" = Table.RenameColumns(..., {"Inflation", "Inflation Rate"})</pre> Renames the value column and divides by 100 to convert from percentage to decimal (e.g. 17.26 → 0.1726).
7. Filter Egypt & Years	<pre>#"Filtered Rows" = Table.SelectRows(..., each ([Country Name] = "Egypt, Arab Rep."))</pre> Retains only Egypt records for the 2020–2024 study period.
8. Add Date Column	<pre>#"Added Custom" = Table.AddColumn(..., "date", each #date([Years], 1, 1))</pre> Creates a proper Date type column from the year integer for time-intelligence compatibility.

Final Output: A clean Inflation dimension table for Egypt covering 2020–2024, with Inflation Rate stored as a decimal for direct use in DAX measures and KPI calculations.

2. Table: Currency Rate (World Bank Exchange Rate)

Step	Description
1. Import Source File	<pre>Source = Csv.Document(File.Contents("...API_PA.NUS.FCRF_DS2_en_csv_v2.csv"), [Delimiter="," , ...])</pre> Loads the World Bank official exchange rate CSV file (EGP per USD).
2. Remove Metadata Rows	<pre>#"Removed Top Rows" = Table.Skip("#Changed Type", 3)</pre> Removes the World Bank metadata header rows identical to the Inflation table.
3. Unpivot Year Columns	<pre>#"Unpivoted Other Columns" = Table.UnpivotOtherColumns(..., "Attribute", "Value")</pre> Pivots from wide format (one column per year) to long format with Years and Exchange Rate columns.
4. Rename Column	<pre>#"Renamed Columns1" = Table.RenameColumns(..., {"Exchange Rate", "EGP per USD"})</pre> Corrects the source column name typo and renames to 'EGP per USD' for clarity.
5. Filter Egypt & Date Range	<pre>#"Filtered Rows2" = Table.SelectRows(... ([date] = #date(2020,1,1) or ... #date(2024,1,1)))</pre> Filters to Egypt only and restricts rows to years 2020–2024.
6. Add Date Column	<pre>#"Added Custom" = Table.AddColumn(..., "date", each #date([Years], 1, 1))</pre> Constructs a date-typed column for time-intelligence alignment with the Master Calendar.

Final Output: A clean Currency Rate dimension table providing annual EGP/USD rates for Egypt from 2020 to 2024, enabling FX Devaluation YoY% calculations.

3. Table: Income Per Capita (World Bank GDP)

Step	Description
1. Import Excel File	<pre>Source = Excel.Workbook(File.Contents("...API_NY.GDP.PCAP.CN_DS2_en_excel_v2.xls"), null, true)</pre> Loads the World Bank GDP per capita (current local currency) Excel file.
2. Navigate to Data Sheet	<pre>#"Navigation 1" = Source{[Name = "Data"]}[Data]</pre> Selects the 'Data' sheet containing the time-series values.
3. Remove Top Rows & Promote Headers	<pre>#"Removed Top Rows" = Table.Skip(#"Changed column type", 2)</pre> Skips 2 pre-header rows. Headers are then promoted in a subsequent step.
4. Unpivot Year Columns	<pre>#"Unpivoted Other Columns" = Table.UnpivotOtherColumns(..., "Attribute", "Value")</pre> Converts the wide year columns into long format with Year and Value columns.
5. Filter Egypt & Years	<pre>#"Filtered Rows" = Table.SelectRows(... [Country Name] = "Egypt, Arab Rep.")</pre> Restricts to Egypt rows and the 2020–2024 date range.
6. Cast to Currency Type	<pre>#"Changed Type1" = Table.TransformColumnTypes(..., {{"Value", Currency.Type}})</pre> Ensures the income value column is stored as a currency/decimal type for accurate monetary calculations.
7. Add Date Column & Reorder	<pre>#"Added Custom" = Table.AddColumn(... "date", each #date([Year], 1, 1))</pre> Generates a properly typed date column and reorders for consistency with other dimension tables.

Final Output: A clean income per capita table for Egypt (2020–2024) in local currency, used to derive Monthly Income EGP by dividing the annual value by 12.

4. Table: Trade_DetailedTradeMatrix (FAO Trade Data)

Step	Description
1. Import Source File	<pre>Source = Csv.Document(File.Contents("...Trade_DetailedTradeMatrix_E_All_Data.csv"), [Delimiter=";", Columns=90, ...])</pre> Loads the FAO Detailed Trade Matrix CSV containing import/export data for Egypt.
2. Remove Flag Columns	<pre>#"Removed Columns" = Table.RemoveColumns(... {"Y1986F", "Y1987F", ..., "Y2024F"})</pre> Removes all flag columns (YearF) and pre-2000 year columns, retaining only relevant year values and metadata.
3. Unpivot	<pre>#"Unpivoted Other Columns" = Table.UnpivotOtherColumns(... "Attribute", "Value")</pre>

Year Columns	Pivots year columns (Y2000–Y2024) to long format with Years and Value columns.
4. Filter Egypt & Products	<pre>#"Filtered Rows" = Table.SelectRows(... [Reporter Countries] = "Egypt")</pre> Retains only Egypt as the reporter country and filters to the 13 selected food commodities.
5. Strip Year Prefix	<pre>#"Replaced Value" = Table.ReplaceValue(... "Y", "", Replacer.ReplaceText, {"Years"})</pre> Removes the 'Y' prefix from year values (e.g. 'Y2023' → '2023') before casting to integer.
6. Add Date Column & Filter Range	<pre>#"Added Custom" = Table.AddColumn(... "Custom", each #date([Years], 1, 1))</pre> Creates a date column. Final filter retains rows for 2020–2024 only.

Final Output: A long-format fact table of Egypt's food trade data (2020–2024), including import quantities and values by partner country and commodity, ready for use in sovereign trade vulnerability analysis.

5. Table: *wfp_food_prices_egy* (WFP Market Price Data)

Step	Description
1. Import Source File	<pre>Source = Csv.Document(File.Contents("...wfp_food_prices_egy.csv"), [Delimiter=",", Columns=16, ...])</pre> Loads the WFP Egypt food prices CSV covering market-level commodity prices across Egyptian governorates.
2. Remove Irrelevant Columns	<pre>#"Removed Columns" = Table.RemoveColumns(... {"market_id", "latitude", "longitude"})</pre> Drops geolocation and ID columns not needed for price analysis.
3. Rename Key Columns	<pre>#"Renamed Columns" = Table.RenameColumns(... {"price", "EGPprice"}, {"commodity", "Product"})</pre> Renames columns to clearer names: EGPprice for local currency price, Product for commodity name.
4. Extract Year & Group	<pre>#"Inserted Year" = Table.AddColumn(... "Year", each Date.Year([date]), Int64.Type)</pre> Extracts year from date then groups rows by Year, market, category, and Product to get average annual prices.
5. Remove Post-2024 Rows	<pre>#"Filtered Rows1" = Table.SelectRows(... [Year] <> 2025 and [Year] <> 2026)</pre> Excludes any partial-year future data to maintain 2020–2024 integrity.
6. Cast Price Columns	<pre>#"Changed Type" = Table.TransformColumnTypes(... {"EGP_price", Currency.Type}, {"USD_price", Currency.Type})</pre> Ensures both price columns are stored as currency type for accurate monetary aggregation.

Final Output: A grouped, clean fact table of average annual food prices per market, category, and product for Egypt from 2020 to 2024, forming the foundation for price trend and affordability analysis.

6. Table: Minimum Wage (Embedded Reference Data)

Step	Description
1. Embedded Binary Source	<pre>Source = Table.FromRows(Json.Document(Binary.Decompress(Binary.FromText(...), Compression.Deflate)))</pre> Decodes a Base64-compressed inline binary containing Egypt's annual minimum wage and crisis stage classifications.
2. Apply Data Types	<pre>#"Changed column type" = Table.TransformColumnTypes(... {"Year", Int64.Type}, {"Monthly_Minimum_Wage_EGP", Int64.Type}))</pre> Casts Year and wage columns to integer types.
3. Add Date Column	<pre>#"Added Custom" = Table.AddColumn(... "date", each #date([Year], 1, 1))</pre> Creates a date column for time-intelligence joins with the Master Calendar.
4. Remove Year Column	<pre>#"Removed Columns" = Table.RemoveColumns(... {"Year"})</pre> Drops the integer Year column after creating the date equivalent.

Final Output: A compact dimension table holding Egypt's monthly minimum wage values and crisis stage labels for each year from 2020 to 2024.

7. Table: FAOSTAT_data (FAO Food Inflation by Category)

Step	Description
1. Import CSV	<pre>Source = Csv.Document(File.Contents("...FAOSTAT_data_en_6-14- 2026.csv"), [Delimiter=",", Columns=17, Encoding=65001])</pre> Loads the FAO food inflation dataset for Egypt, providing inflation rates broken down by food item category.
2. Remove Unnecessary Columns	<pre>#"Removed Columns" = Table.RemoveColumns(... {"Domain Code", "Area Code (M49)", "Year Code", ...})</pre> Eliminates metadata, code, and flag columns not required for analysis.
3. Divide Value by 100	<pre>#"Divided Column" = Table.TransformColumns(... {"Value", each _ / 100}))</pre> Converts percentage values (e.g. 31.9) to decimal form (0.319) for consistency with other rate measures.
4. Group by Year, Area, Item	<pre>#"Grouped Rows" = Table.Group(... {"food inflation %", each List.Average([Value])}))</pre> Averages monthly inflation values within each year-item-element group to produce annual averages.
5. Add Date Column	<pre>#"Added Custom" = Table.AddColumn(... "Custom", each #date([Year], 1, 1))</pre> Creates the standard date column used for joining with the Master Calendar table.

Final Output: A clean dimension table of average annual food inflation rates by commodity category for Egypt (2020–2024), used to compute the Avg Food Inflation % KPI and sectoral transmission visuals.

Chapter 2: System Analysis & Design

- ❖ Database Design & Data Modeling
 - ❖ UI/UX Design & Prototyping
 - ❖ Dashboard Navigation Guide

Chapter 2: System Analysis & Design

System Analysis & Design

1. Database Design & Data Modeling

Data Schema Diagram

The analytical data model in Power BI is designed using a Star Schema with elements of a Snowflake Schema, where multiple fact tables share common dimension tables linked through a Master Calendar. This design supports model purity, consistent granularity, and efficient filtering performance across the macroeconomic and food security analytical workloads.

A. Model Structure and Fact Separation

To maintain semantic clarity and ensure accurate analytical calculations, the model organizes operational metrics into two categories:

- **Fact Tables:** wfp_food_prices_egy (core food price fact table) and Trade_DetailedTradeMatrix (trade flow fact table). These contain additive numeric measures and link to dimensions via date and product keys.
- **Dimension Tables:** INFLATION, Currency Rate, Income Per Capita, Minimum Wage, and FAOSTAT_data. These provide filtering context and lookup values for the fact tables.
- **Justification:** Separating facts from dimensions enforces Measure Purity, prevents metric duplication, and enables reliable computation of composite KPIs such as the Income Erosion Index ($\text{Avg Local Price} \times 30 / \text{Minimum Wage}$) and Simulated Price under currency shock scenarios.

B. Master Calendar — The Spine of Time Intelligence

A Master Calendar table was created to serve as the single date dimension, enabling consistent YoY comparisons across all tables. All source tables were aligned to this calendar via their date columns, ensuring that DAX time-intelligence functions (PREVIOUSYEAR, DATEADD, CALCULATE) operate correctly across every measure.

C. Data Schema Overview

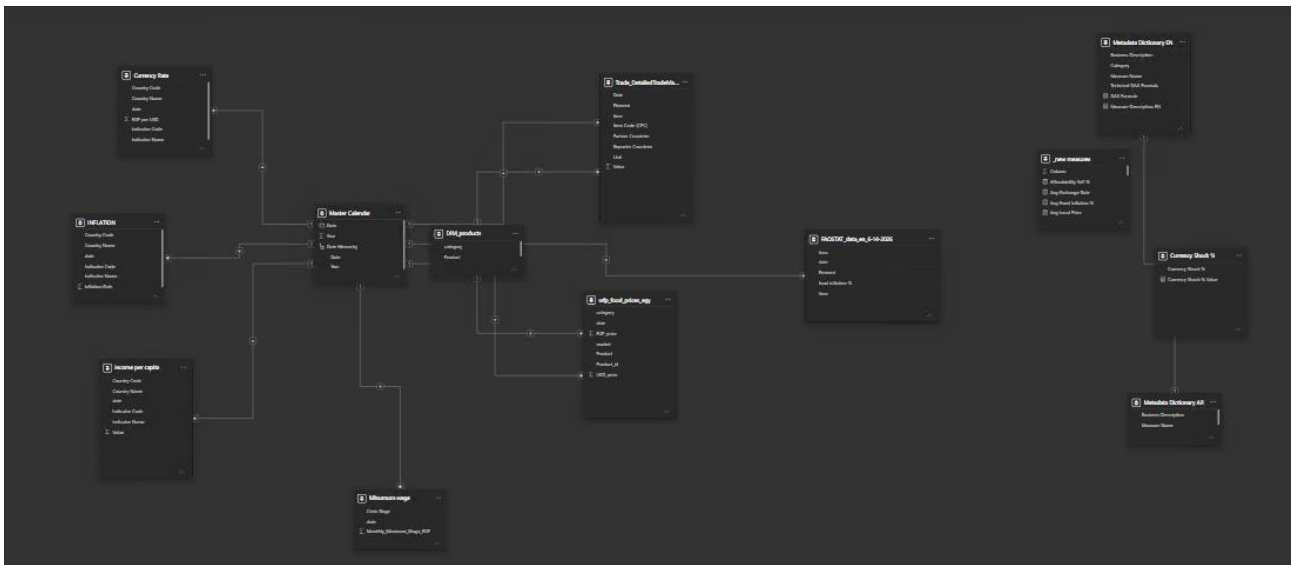
The following table summarizes all tables in the data model, their type, primary relationship key, and key columns:

Table Name	Type	Primary Key	Key Columns
INFLATION	<i>Dimension</i>	date	Country Name, Country Code, Inflation Rate, date
Currency Rate	<i>Dimension</i>	date	Country Name, EGP per USD, date
wfp_food_prices_egy	<i>Fact</i>	Product_id	Year, market, category, Product, EGP_price, USD_price

Trade Matrix	Fact	Date	Reporter Countries, Partner Countries, Item, Element, Value, Date
income per capita	Dimension	date	Country Name, Value (EGP), date
Minimum Wage	Dimension	date	Monthly_Minimum_Wage_EGP, Crisis Stage, date
FAOSTAT_data	Dimension	date	Area, Item, Element, food inflation %, date
Master Calendar	Date Dimension	date	date, Year, Month, Quarter

Figure 4-2-1 Data Model Schema Overview

The following placeholder represents the Power BI data model diagram to be inserted:



2. UI/UX Design & Prototyping

This section outlines the UI/UX design system, layout strategy, interaction flow, and visual identity used in the Dashboard. The purpose is to ensure a consistent, scalable, and user-centered design across all dashboard pages.

2.0 Design Tools & Methodology

A standardized UI/UX workflow was followed throughout the dashboard development process. Initial wireframes were created to validate layout logic and user flow. These structures were then translated into high-fidelity visual mockups using Figma, applying the approved color palette, typography, and interaction guidelines. The finalized layouts were subsequently implemented in Microsoft Power BI, ensuring full alignment between design intent and functional execution.

Tools Used:

- ☐ Figma: Visual design & component styling
- ☐ Draw.io: Wireframes & structural mapping
- ☐ Power BI: Final dashboards & interactive implementation

2.1 Wireframes & Mockups

The following wireframe descriptions represent the structure of the dashboard pages. Actual visuals are implemented in Power BI.

- Home Page: Title, introduction, shortcuts, and navigation.
- Page 1 – The Macro Shock Catalyst: Macroeconomic overview with CPI, exchange rate, and FX devaluation trends.
- Page 2 – Sectoral Transmission: Food category analysis, risk status matrix, and key influencers.
- Page 3 – Household Reality (Drill-Through): Wage vs. inflation erosion, affordability index, income erosion gauge.
- Page 4 – Sovereign Trade Vulnerabilities: Trade partner map, import cost analysis, and price simulation.
- Tooltip Pages: Contextual micro-dashboards on hover for Page 1 (economy baseline) and Page 4 (trade by year).

Home	
Page 1: Macro Shock Catalyst	Page 2: Sectoral Transmission
Page 3 (Drill-Through): Household Reality	Page 4: Sovereign Trade Vulnerabilities

Figure 4-4-1-1 Dashboard Navigation Map

2.3 UI/UX Guidelines

Color Palette

The dashboard uses a dark industrial theme inspired by a deep-navy and midnight blue palette, accented with teal highlights to convey financial gravity and analytical precision.

Color	Hex Code	Usage
White	#FFFFFF	Background / Cards
Gray	#838383, #CCCCCC	Strokes & Borders
Soft Black (Navy)	#0D0D0D / #1F3864	Navigation Panel, Headers
Teal Accent	#2E5FA3 / #2E75B6	KPIs, Chart Fills & Highlights
Gold Accent	#C09A2F	Emphasis Values & Anomalies

Typography

The chosen fonts prioritize clarity and analytical hierarchy across KPIs, titles, and body visuals.

Element	Font Style
Headers / Page Titles	Segoe UI Bold / Light — 18–24 pt
KPIs & Metrics	Segoe UI Semibold — 24–36 pt

Body Text	Segoe UI Regular — 12–14 pt
Captions / Notes	Segoe UI Light — 10–11 pt

Interaction Flow

- Hovering over chart elements reveals tooltip micro-dashboards with economy baseline and trade breakdown context.
- Selecting a year via the timeline slicer refreshes all visuals contextually across all pages.
- Right-click drill-through from Page 1 or Page 2 navigates to the Household Reality detail page.
- The navigation panel remains static for quick switching between all main pages.
- The Currency Shock slicer on Page 4 dynamically updates the Simulated Price line chart in real time.

Design Decisions & Rationale

The design is built around analytical readability and narrative clarity. The dark theme reduces visual noise and draws the eye to the data. The teal and navy palette was chosen to evoke financial intelligence and institutional gravity — reflecting the seriousness of Egypt's food security challenge. Horizontal bar charts and area charts were selected to make YoY trend comparison as immediate as possible for the reader.

3. Dashboard Navigation Guide

3.1 Page 1: The Macro Shock Catalyst

The Macro Shock Catalyst page serves as the entry point to the analysis, providing the macroeconomic context that underpins the food security crisis. It is the first page a user lands on after the home screen.

The four KPI cards at the top of the page summarize Egypt's macro environment. A 2020–2024 year range slicer in the top-right allows the user to isolate any specific year or span. Each card drives the four charts below it.

Visuals:

- Line and Clustered Column Chart — Headline CPI and Avg Exchange Rate by Year: Overlays the inflation rate (line) against the nominal exchange rate (bars) to visualize how currency depreciation aligns with inflation surges.
- Area Chart — FX Devaluation YoY% by Year: Shows the annual percentage devaluation of the EGP against the USD, with peaks at +59.84% in 2023 revealing the sharpest depreciation episode.
- Line Chart — Headline CPI by Year: Plots annual CPI trajectory from 5.04% in 2020 to 33.88% in 2023, providing a clear trendline for the inflation escalation narrative.
- Scatter Chart — Headline CPI and Avg Food Inflation% by Date: Plots each year as a dot in two-dimensional space, exposing the structural gap where food inflation consistently outpaces headline CPI.



Figure 3-1-1 Dashboard Page 1: The Macro Shock Catalyst

3.2 Page 2: Architecture — Sectoral Transmission (Food vs. Economy)

The Sectoral Transmission page drills into the food category level, revealing how macroeconomic shocks transmit unevenly across different food product groups — from cereals to oils and dairy.

Visuals:

- Clustered Column Chart — Price YoY% and Avg Food Inflation% by Product: Compares individual product price growth against the overall food inflation benchmark for each product, identifying which items exceed the average.
- Matrix — Product vs Year (Price PY, Risk Status): A heatmap-style matrix showing prior-year prices and a color-coded risk status (STABLE / WARNING / CRITICAL) for each product across years, using the Affordability YoY% measure.
- Key Influencers Analysis — Price YoY%: An AI-powered visual from Power BI that identifies the top factors explaining price increases, including category type and EGP/USD rate changes.
- Ribbon Chart — Price YoY% by Year and Category: Tracks how food category rankings shift year over year, showing which categories (oils and fats, cereals) rose most dramatically during crisis years.



Figure 3-2-1 Dashboard Page 2: Sectoral Transmission (Food vs. Economy)

3.3 Page 3 (Drill-Through): Household Reality — Wages vs. Real Erosion

The Household Reality page is a drill-through destination accessible from Page 1 or Page 2. It anchors the macroeconomic data to human impact — showing how wage growth has failed to keep pace with rising food prices, eroding real purchasing power for Egyptian households.

Visuals:

- Area Chart — Average Monthly Minimum Wage and Monthly Income EGP by Year: Dual-axis chart overlaying wage growth against income per capita trends, showing the widening gap between official wage floors and actual income levels.
- Line Chart — Units Affordable by Year: Tracks how many units of a reference food basket a minimum-wage household could afford each year, peaking in 2022 at ~400 units and declining to ~280 units by 2024.
- Gauge — Income Erosion Index: A gauge visual displaying the proportion of monthly minimum wage consumed by a month's food basket, with a current reading of \$0.1852 out of a \$0.37 scale.
- Matrix — Income Erosion Index by Product and Year: Shows the income erosion index for sugar (the tracked product) across all five years, making it easy to see how affordability deteriorated as the EGP weakened.



Figure 3-3-1 Dashboard Page 3 (Drill-Through): Household Reality

3.4 Page 4: Sovereign Trade Vulnerabilities

The Sovereign Trade Vulnerabilities page examines Egypt's external exposure — how dependency on specific trade partners and the weakening EGP create compounding import cost risks for critical food commodities.

Visuals:

- Azure Map — Import Volume (MT) by Partner Countries: A bubble map scaled by import volume showing geographic trade concentration, with the largest bubbles in Eastern Europe and North America (primary wheat and grain sources).
- Line and Clustered Column Chart — Import Value (USD) and Total Import Cost (EGP) by Year: Overlays USD-denominated import values (relatively stable) against EGP-denominated total costs (steeply rising), illustrating how currency depreciation amplifies the local cost of imports.
- Decomposition Tree — Total Import Cost (EGP): An expandable tree that breaks down total import cost by product category, allowing users to identify which commodities drive the highest EGP expenditure.
- Line Chart — Avg Local Price and Simulated Price (EGP) by Year: Compares actual observed local prices against a what-if simulated price under a user-defined currency shock scenario, powered by the Currency Shock % slicer.



Figure 3-4-1 Dashboard Page 4: Sovereign Trade Vulnerabilities

3.5 Tooltip Pages (Pages 5 & 6)

Two tooltip micro-pages provide on-hover contextual details without requiring navigation:

- Page 5 (Tooltip for Page 1): Displays three KPI cards — General Economy Baseline, Caloric Survival Cost, and the Survival Gap — giving an instant economic health snapshot when hovering over Page 1 visuals.
- Page 6 (Tooltip for Page 4): Displays a Line and Clustered Column Chart showing Import Volume (MT) and Import Value (USD) by Year, giving trade context on hover over the Page 4 map.

4. DAX Measures — Analytical Engine

The following custom DAX measures were developed to power the KPIs and visuals across all dashboard pages. They are organized by analytical theme.

4.1 Affordability Nexus Measures

Measure Name	Purpose / Logic
Affordability YoY %	YoY change in Units Affordable. Uses PREVIOUSYEAR on Master Calendar date.
Food Crisis Premium	Avg Food Inflation % minus Headline CPI. Measures how much food inflation exceeds general inflation.
Income Erosion Index	Avg Local Price × 30 divided by monthly minimum wage. Represents the basket cost-to-wage ratio.
Price vs Inflation Gap	Price YoY % minus Headline CPI. Shows how food price growth differs from overall inflation.
Real Purchasing Power Change	Difference between current and prior-year Avg Local Price, showing real price movement.

Simulated Price (EGP)	Avg Local Price × (1 + Currency Shock %). Enables scenario modeling under FX shocks.
Units Affordable	Monthly Income EGP divided by Avg Local Price. Measures household purchasing capacity.

4.2 Core Measures

Measure Name	Purpose / Logic
Avg Exchange Rate	AVERAGE of EGP per USD from the Currency Rate table.
Avg Food Inflation %	AVERAGE of food inflation % from the FAOSTAT dataset.
Avg Local Price	AVERAGE of EGP_price from the WFP food prices table.
Headline CPI	AVERAGE of Inflation Rate from the World Bank CPI table.
Monthly Income EGP	Annual income per capita divided by 12, using DIVIDE for safe division.

4.3 Prescriptive Analytics Measures

Measure Name	Purpose / Logic
FX Victim Flag	Returns 1 if FX Devaluation YoY% > 15% AND Price YoY% > 20%, flagging products critically impacted by currency shock.
Max Category Spike %	MAXX across food categories of Price YoY%, identifying the single highest-spiking category in any year.
Most Volatile Category	TOPN(1) filter on Price YoY% to return the category name with the highest annual price increase.
Risk Status	SWITCH logic assigning CRITICAL (Affordability YoY% ≤ -20%), WARNING (< 0%), or STABLE (≥ 0%) labels.
Survival Basket Price Growth %	Average Price YoY% across the 5 survival basket products: Rice, Sugar, Wheat flour, Vegetable oil, Poultry.

4.4 Macro Trade Reality Measures

Measure Name	Purpose / Logic
Global Cost Per Ton (USD)	Import Value (USD) divided by Import Volume (MT) using safe DIVIDE.
Highest Risk Partner	TOPN(1) filter on Import Quantity to identify the single largest trade partner by volume.
Import Value (USD)	SUM of trade matrix values filtered to Element = 'Import Value', multiplied by 1000 for unit conversion.
Import Volume (MT)	SUM of trade matrix values filtered to Element = 'Import Quantity'.
Total Import Cost (EGP)	Import Value (USD) × Avg Exchange Rate, converting USD cost to local EGP equivalent.

4.5 Time Intelligence Measures

Measure Name	Purpose / Logic
FX Devaluation YoY %	Year-over-year change in Avg Exchange Rate using PREVIOUSYEAR on Master Calendar date.
Price PY	Prior-year Avg Local Price using PREVIOUSYEAR — the baseline for Price YoY% calculation.
Price YoY %	DIVIDE(Avg Local Price – Price PY, Price PY). Annual percentage price change for each product.

Chapter 3: Analysis & Insights

- ❖ Insights
- ❖ Target
- ❖ Recommendations

Chapter 3: Analysis & Insights

This chapter presents the analytical findings, key insights, and strategic recommendations derived from the full macro shock analysis. The content draws on four analytical lenses: macroeconomic structure, product-level supply chain volatility, household purchasing power, and geopolitical trade exposure. All figures are sourced directly from the validated Power BI dashboard.

Section I: The Structural Shock — Macroeconomic Disintegration (2020–2024)

Between 2020 and 2024, Egypt’s macroeconomic architecture underwent a structural transformation that cannot be characterised as a temporary downturn. The data reveals a permanent recalibration of cost baselines, driven by a sovereign currency collapse that amplified every external and domestic price signal.

A. Key Performance Indicators: The Macroeconomic Dashboard

KPI 1 | Headline CPI — The New Inflation Floor

Metric	Value
2020 Baseline	5.04%
2023 Peak	33.88%
2024 Rate	28.27%

Insight:

The modest deceleration from 33.88% to 28.27% must not be misread as recovery. The economy has established a new, elevated inflation floor rather than a temporary spike. The 2020 baseline of 5.04% is structurally unreachable without a sovereign intervention of exceptional magnitude.

Recommended Action

Incorporate a rolling inflation indexation multiplier into all long-term financial forecasts. Reliance on historical fixed baselines from the pre-2022 era will produce systematically underestimated cost projections.

KPI 2 | Average Exchange Rate — The Purchasing Power Erosion Engine

Metric	Value
2020 Average Rate	15.76 EGP / USD
2024 Average Rate	45.30 EGP / USD

Trajectory	Unbroken upward — zero consolidation years
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Insight:

The persistent upward trajectory from 15.76 to 45.30 EGP/USD represents an unrelenting erosion of purchasing power for all imported goods. This is not cyclical volatility — it is a permanent structural alteration of import cost architecture.

Recommended Action

Reallocate operational budgets toward local procurement contracts to limit future exposure to severe currency fluctuations. The risk premium on imported inputs is now structurally embedded and must be treated as a permanent cost line.

KPI 3 | Max FX Devaluation YoY% — The Velocity of Collapse

Year	Devaluation
2020	0.00%
2022	22.47%
2023 (Peak)	59.84%
2024	47.91%

Insight:

The sheer velocity of the currency's decline — 59.84% in 2023 alone — acted as the primary catalyst for immediate market instability and supply chain disruption. Critically, 2024's 47.91% rate confirms that devaluation pressure has not dissipated; it has merely shifted registers.

Recommended Action

Implement automated price triggers that adjust margins the moment FX devaluation exceeds a predefined 10% threshold. Reactive pricing in a 47–59% devaluation environment is commercially catastrophic.

KPI 4 | Average Food Inflation % — The Decoupled Crisis

Metric	Value
2020 Baseline	-0.15% (deflation)
2023 Peak	63.57%
2024 Rate	33.82%

Insight:

Food inflation surged from a negative baseline of -0.15% in 2020 to a devastating 63.57% in 2023 — nearly double the headline CPI peak. This decoupling exposes extreme volatility and fragility specifically within the food supply chain, independent of broader macroeconomic dynamics.

Recommended Action

Shift strategic sourcing focus to staples with high historical price elasticity. Supply chains capable of absorbing sudden inflation shocks must be prioritised above all standard procurement criteria.

B. Visual Analytics: Reading the Structural Shock

Visual 1 | Headline CPI vs. Average Exchange Rate: The exchange rate’s ascent to 45.30 tracks tightly against the Headline CPI trajectory culminating at 28.27% in 2024, proving that currency depreciation is the absolute primary driver of domestic inflation. Use the FX rate as a direct leading indicator — secure raw material volumes preemptively based on anticipated currency adjustments.

Visual 2 | FX Devaluation YoY%: The chart cascades from 0.00% in 2020 through compounding shocks: 22.47% in 2022, 59.84% in 2023, and 47.91% in 2024. There have been zero years of currency recovery or consolidation across the entire five-year period.

Visual 3 | Food Inflation vs. Headline CPI: Food inflation decoupled entirely from general CPI starting in 2022, spiking nearly twice as high during the crisis apex at 63.57% vs 33.88%. This divergence signals a structural fragility in food-specific supply chains that persists independently of broader monetary conditions.

Section II: The Supply Chain Fracture — Product-Level Volatility & the Survival Basket

The Survival Basket — a curated index of essential caloric commodities — became the most accurate barometer of market stress between 2021 and 2024. The data reveals not a uniform price increase, but a cascading, rotating crisis that migrated across food categories systematically.

A. Key Performance Indicators: The Survival Basket Intelligence

KPI 1 | Survival Basket Price Growth — The Cost of Survival

Metric	Value
2023 Basket Growth	86.29%
2024 Basket Growth	29.48% (cumulative on elevated base)

Insight:

The 2024 deceleration to 29.48% represents growth compounding onto an already cripplingly high baseline — not a restoration of affordability. The cost floor has been reset at a permanently elevated level.

Recommended Action

Position survival basket commodities as heavily subsidised or loss-leader items to maintain essential market footfall. Any distributor that fails to absorb this structural premium will experience severe consumer attrition at the lowest income tiers.

KPI 2 | Max Category Spike % — The Near-Doubling Event

Metric	Value
2023 Maximum Spike	96.60%
2024 Maximum Spike	57.94%

Insight:

Specific commodity categories experienced near-100% price doubling within a single calendar year. Standard cost-plus pricing models built on 5–15% annual increase assumptions are rendered entirely obsolete.

Recommended Action

Mandate daily inventory monitoring for any category showing year-over-year spikes exceeding 40%. Prevention of sudden capital drains requires real-time surveillance, not retrospective CPI-based reporting cycles.

KPI 3 | Most Volatile Category — The Rotating Crisis

Year	Most Volatile Category
2021	Oils and Fats
2022	Vegetables and Fruits
2023	Miscellaneous Food
2024	Miscellaneous Food (continued)

Insight:

Market volatility aggressively shifts between core food groups annually. The annual rotation of the ‘most vulnerable’ category is itself the signal — it indicates that the pressure is systemic and structural, not commodity-specific.

Recommended Action

Diversify category sourcing geographically so that a geopolitical or climatic shock to one food category does not paralyse the entire supply line. Single-origin category dependence is a material enterprise risk in this environment.

KPI 4 | Critical Commodities Count — The Crisis Breadth Signal

Year	Critical Items
2022	1 commodity

2023	9 commodities
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Insight:

The crisis broadened with devastating speed in 2023, infecting nearly the entirety of the survival basket. The jump from 1 to 9 critical items in a single year demonstrates a systemic contagion mechanism, not isolated supply failures.

Recommended Action

Automatically trigger emergency strategic reserve releases when the critical commodity count exceeds three items simultaneously. A threshold-based response protocol prevents the 1→9 escalation pattern from repeating.

B. Visual Analytics: Mapping the Product-Level Crisis

The 2023 data reveals a systemic failure cascade: Sugar spiked 96.60%, Meat/Chicken 94.03%, and Eggs 87.85% — triggering a near-total CRITICAL designation across core staples. Entering 2024, Potatoes (62.25%), Sugar (57.94%), and Sunflower Oil (34.63%) remain under WARNING status. The ribbon chart confirms that Sugar demonstrates the highest resistance to price correction, sustaining near-peak pressure across consecutive years as the only commodity to do so.

Section III: The Household Reality — Purchasing Power Collapse & the Wage Illusion

The macroeconomic and supply chain data ultimately converges on a single human outcome: the systematic erosion of the Egyptian citizen’s purchasing power. The findings mathematically dismantle three persistent policy narratives — that wage increases restore prosperity, that nominal growth reflects real wealth, and that current income ratios are sustainable.

A. The Volatile Recovery Trap — Purchasing Power YoY%

Year	Purchasing Power Change
2021	+4%
2022	-12%
2023	-32%
2024	+2% (statistically negligible)

Insight:

The data mathematically disproves the narrative of economic recovery. After a baseline improvement of +4% in 2021, purchasing power suffered a cascading collapse of -12% in 2022 and a devastating -32% in 2023. The +2% growth recorded in 2024 is statistically negligible — it proves that massive nominal wage hikes act merely as a localised inflationary adjustment, failing to generate true wealth accumulation for the citizen.

B. The Decoupling of Wages and Wealth — The Illusion of Prosperity

Metric	Value
2021 Minimum Wage	2,400 EGP → 3,056 units affordable
2024 Minimum Wage	6,000 EGP → 1,853 units affordable
Nominal Wage Change	+150%
Physical Purchasing Power Change	-39.35%

Insight: The Illusion of Wealth

In 2021, a 2,400 EGP minimum wage secured 3,056 physical units. By 2024, the state doubled the wage to 6,000 EGP, yet the worker can now only afford 1,853 units. While nominal wages surged by 150%, physical purchasing power collapsed by 39.35%. The citizen holds more paper money but commands less actual food.

C. The Wage-Push Inflation Risk — Compressing the Income Gap

Year	Min Wage as % of Avg Income
2023	3,000 EGP / 7,388 EGP = 40.6%
2024	6,000 EGP / 9,941 EGP = 60.3%
One-Year Ratio Shift	+19.7 percentage points

Insight:

The ratio of minimum wage to average income violently spiked from 40.6% to 60.3% in a single year. This rapid, mandated wage-push creates a high risk of a wage-price spiral, as businesses immediately raise prices to cover the sudden 20% jump in relative labour costs.

D. The Structural Deficit Warning — The 2.76% Erosion Signal

Metric	Value
Income Erosion Rate (Single Staple)	2.76% of average monthly income
Retained Income (Single Staple)	97.24%
Projected Basket Erosion (10–12 items)	~35% of total income consumed

Insight:

A single staple commodity currently devours 2.76% of the average income. When multiplied across a standard Survival Basket containing 10 to 12 essential items, the Egyptian consumer is immediately pushed into a structural deficit. Essential caloric intake alone will consume upwards of 35% of total income, leaving virtually nothing for housing, healthcare, or education.

Section IV: The Grand Synthesis — Geopolitical Exposure & the Currency Multiplier

Section IV elevates the analysis to its structural root cause: the intersection of a collapsed sovereign currency and a dangerously concentrated geopolitical import footprint. These two forces interact in a multiplier relationship that transforms manageable global price movements into catastrophic domestic cost explosions.

A. The Currency Multiplier Effect

Metric	Value
Import Cost 2020 (USD)	\$374 Million
Import Cost 2024 (USD)	\$1.23 Billion
Real Import Growth (USD)	~3.3x
Import Cost 2020 (EGP)	5.89 Billion EGP
Import Cost 2024 (EGP)	55.77 Billion EGP
EGP Budget Explosion	~9.5x

Insight:

Egypt is not a victim of a global food shortage — it is a victim of an exchange rate collapse. A 3.3x real USD import cost increase translated into a 9.5x EGP budget explosion. The Egyptian Pound's devaluation acts as an 'inflation multiplier,' proving that Egypt's food inflation is a domestic, sovereign-driven crisis.

B. The Black Sea Trap — Critical Geopolitical Exposure

Trade Partner	Volume (MT)
Total Tracked Imports	2.68 Million MT
Russian Federation	1,033,453 MT (~38%)
Ukraine	277,551 MT (~10%)
Combined Black Sea Dependency	~1.31 Million MT (~49%)

Insight:

Almost 49% of Egypt's essential survival basket is tied to an active, prolonged war zone. A single escalation in the Black Sea supply chain will instantly paralyse half of Egypt's national food reserves, irrespective of the Central Bank's reserve position.

C. The Oil Haemorrhage — The Single Largest Structural Vulnerability

Commodity	Import Volume (MT)
Sunflower Oil	1,487,120 MT (>55% of tracked volume)
Potato	822,771 MT
Sugar	310,412 MT
Maize Oil	127,000 MT

Insight:

Sunflower Oil alone accounts for more than 55% of the imported volume. Egypt is relying on Russian and Ukrainian exports specifically to cover this massive 1.48 million MT oil deficit — a concentrated vulnerability with a direct line to the most unstable supply region on the planet.

D. The Ticking Time Bomb — FX Shock Simulator

Scenario	Price per Unit (EGP)
Survival Basket Avg Price (2021)	16.08
Survival Basket Avg Price (2024)	45.14
Simulated Post-Shock Price	58.68
Simulated Price Gap	~13.5 EGP per unit

Insight:

The FX simulator proves mathematically that a further currency shock will instantly spike the survival basket price from 45.14 EGP to 58.68 EGP. The ~13.5 EGP per unit jump is entirely driven by FX expectations, not actual consumer demand. Without preemptive subsidies, the 1,853 units currently affordable will instantly collapse.

Section V: The Master Action Plan — Integrated Policy Interventions

The following recommendations are synthesised from the full analytical corpus. Each intervention is grounded in specific, verified numerical evidence and targeted at the precise mechanism identified as the source of the corresponding market failure.

Intervention 1.1 | Automated FX Price Triggers

The Evidence

FX devaluation hit 59.84% in 2023 and 47.91% in 2024, with zero recovery years across the five-year period.

The Action

Implement automated price triggers that adjust commercial and procurement margins the moment FX devaluation exceeds a predefined 10% threshold. Reactive pricing in an environment with 47–59% devaluation velocity is commercially untenable.

Intervention 1.2 | Establish the 13.5 EGP Preemptive Subsidy Buffer

The Evidence

The FX simulator shows a looming 13.5 EGP per unit gap, with the survival basket price projected to spike from 45.14 EGP to 58.68 EGP upon the next currency shock.

The Action

The Ministry of Finance must establish a ring-fenced FX Shock Buffer Fund. The state needs to prepare an emergency in-kind subsidy equivalent to 13.5 EGP per essential unit, automatically triggered the moment the Central Bank adjusts the exchange rate.

Intervention 2.1 | Enforce a 20% Geopolitical Sovereign Cap

The Evidence

Russia supplies ~38% of Egypt's import volume (1,033,453 MT). Combined with Ukraine's 277,551 MT, nearly 49% of essential reserves are tied to an active war zone.

The Action

The Ministry of Supply must mandate a 20% strict sovereign risk cap. No single nation may supply more than 530,000 MT of strategic reserves. Excess Black Sea volume must be redirected to Brazil (92,634 MT) and Canada (85,157 MT), both already within the supply network.

Intervention 2.2 | Launch the 1.5 Million Ton Oil Localisation Initiative**The Evidence**

Egypt imports 1,487,120 MT of Sunflower Oil plus 127,000 MT of Maize Oil at a total EGP import cost of 55.77 billion EGP — over 55% of the tracked import volume.

The Action

The Ministry of Agriculture must launch an aggressive import substitution programme for oilseed crops. Reallocating just 15% (~8.3 billion EGP) of the import budget to local farmer contracts can realistically replace 370,000 MT (25% of the oil deficit) within 18 months.

Intervention 2.3 | Activate Emergency Stockpile Releases at the 3-Item Threshold**The Evidence**

Critical commodity count jumped from 1 item in 2022 to 9 items in 2023 — a 900% breadth expansion in a single year, signalling systemic contagion rather than isolated supply failures.

The Action

Automatically trigger emergency strategic reserve releases when the critical commodity count exceeds three items simultaneously. This circuit-breaker, set conservatively at three, prevents the 1→9 escalation pattern from repeating.

Intervention 3.1 | Pivot to Quantity-Based Subsidies — The 29-Unit Proof**The Evidence**

The 100% nominal wage increase from 3,000 EGP to 6,000 EGP yielded a physical purchasing power recovery of exactly 29 units — from 1,824 to 1,853 units affordable.

The Action

Advise the Ministry of Finance to halt further nominal wage hikes for the next 6 months. Redistribute 50% of the allocated wage-increase budget directly into in-kind physical subsidies. Lowering a commodity's price by 20% increases Units Affordable far more effectively than raising wages by 20%.

Intervention 3.2 | Trigger the Purchasing Power Floor — The 3.5% Rule**The Evidence**

The baseline Income Erosion Index stands at 2.76% of average monthly income for a single staple. A 10–12 item basket pushes total erosion to approximately 35% of income — a structural deficit state.

The Action

The Ministry of Supply must adopt the dashboard as an automated real-time alert system. If the Income Erosion Index for any single staple crosses the 3.5% threshold, it must trigger an automatic mandatory release of strategic stockpiles to break localised monopolies before inflation spreads.

Intervention 3.3 | Implement SME Labour Tax Credits — The 60% Squeeze Response**The Evidence**

The ratio of minimum wage to average income spiked from 40.6% in 2023 to 60.3% in 2024 — a 19.7 percentage point increase in a single year, placing unsustainable burden on SMEs.

The Action

The state must introduce a targeted Wage Credit. Any SME retaining its workforce at the 6,000 EGP minimum wage tier should receive a proportional deduction on corporate tax liabilities to offset the 20% surge in relative labour costs.

Conclusion: The Strategic Imperative

The data assembled across these four analytical dimensions — macroeconomic shock, supply chain fracture, household purchasing power collapse, and geopolitical import concentration — converges on a single, unambiguous conclusion: Egypt's food inflation crisis is not a temporary market disturbance. It is a structural, sovereign-driven emergency with permanent consequences for affordability, social stability, and national food security.

Three structural fault lines define the risk architecture:

- The Currency Multiplier: A 3.3x real USD import cost increase translated into a 9.5x EGP budget explosion — proving that domestic food inflation is fundamentally a currency crisis, not a supply crisis.
- The Geopolitical Concentration: 49% of essential reserves are sourced from an active war zone. No level of financial reserves can insulate against a Black Sea supply chain disruption of this magnitude.
- The Purchasing Power Illusion: A 150% nominal wage increase delivered a 39.35% collapse in physical purchasing power. The citizen is trapped in a mathematically deteriorating position regardless of headline salary figures.

The window for preemptive intervention is narrow.

The FX Shock Simulator projects a further 13.5 EGP per unit price spike upon the next currency adjustment. The 1,853 units currently affordable by the minimum wage earner will collapse the moment the Central Bank acts — unless the 13.5 EGP buffer, the geopolitical diversification mandate, and the quantity-based subsidy pivot are operationalised in advance.

This analysis provides the quantitative foundation for intervention. The data has spoken with precision.